

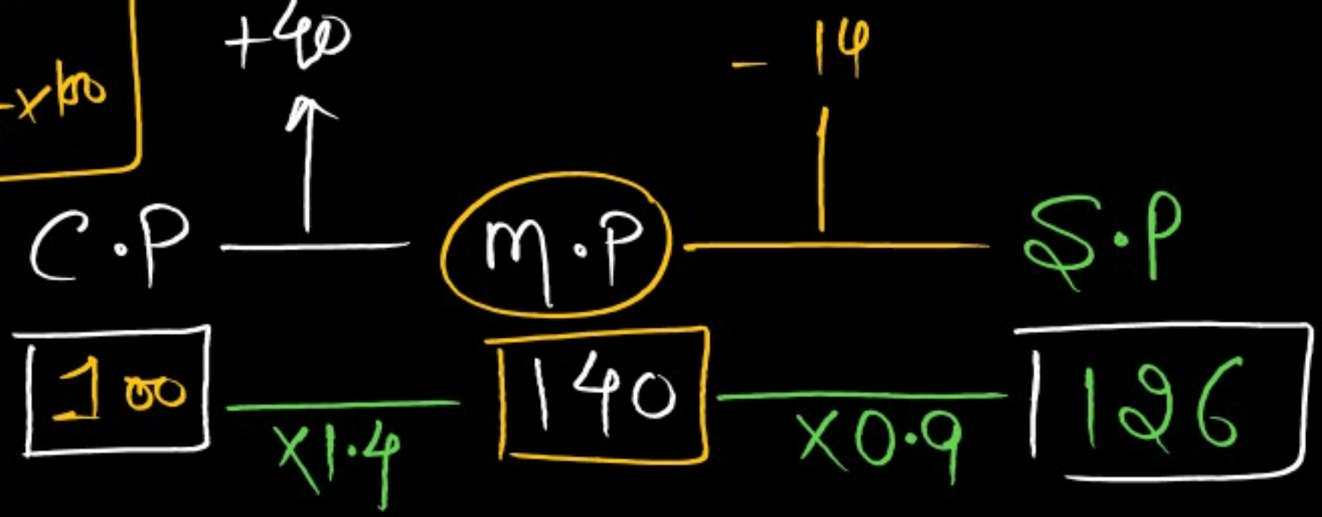
Topic : Discount(P/L)

LIVE CLASS

Discount Concept : Vishal Mega Mart

① $P/L = \frac{CP}{SP}$

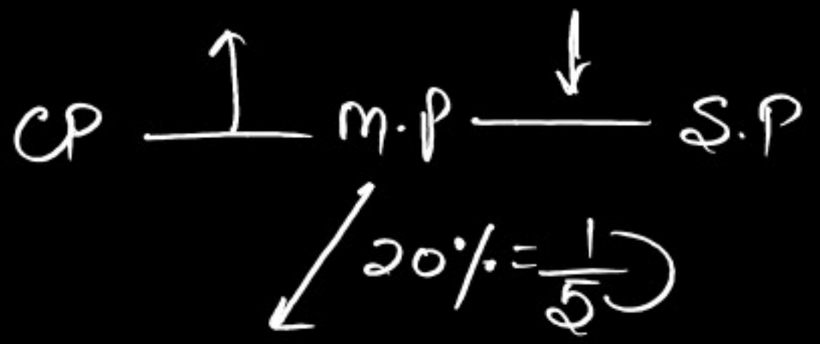
② $D\% = \frac{D}{M.P} \times 100$



$M.P \times M.F = S.P$

Question1 :

The marked price of an immersion rod in an electronic store is Rs.900. The store offers a discount of 12% on its sale. At what price (in Rs.) can a customer buy it from the electronic store?



M.P × M.F = S.P

9

900 × 88 =

7 10

792

M.P = 900

d% = 12%

S.P = ?

Question 2:

A shopkeeper announced 17% rebate on the marked price of an article. If the selling price of the article is Rs.1,245, then the marked price (in Rs.) of the article is :

M.P = 1500

$$M.P \times \frac{83}{100} = 1245$$

15

$$M.P \times M.F = S.P$$

Concept : Successive Discount

$$S = a + b + \frac{ab}{100}$$

$-10 - 10 + 1 = -19\%$

$= -\left(a + b - \frac{ab}{100}\right)$

Direct

$10 + 10 - \frac{100}{100} = 19\%$

① Basic

-10% -10%

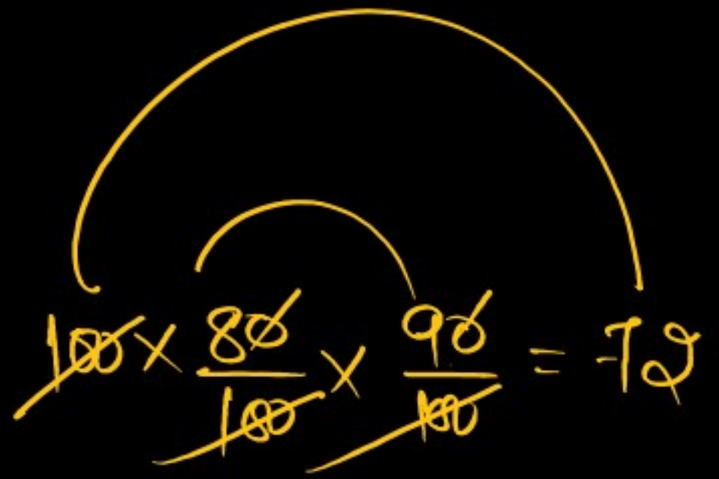
$100 \xrightarrow[-10]{10\%} 90 \xrightarrow[-9]{10\%} 81$

$D = 19\%$

② ~~XX~~

$100 \times \frac{90}{100} \times \frac{90}{100} = 81$

③



Question 3:

A single discount equivalent to two successive discount of 20% and 10% on the marked price of an article is?

$= 20 + 10 - \frac{20}{100}$

$= 28\%$

Question 4:

Two successive discounts of 20% and 20% are equivalent to a single discount of:

[A]34%
[B]36%
[C]40%
[D]42%

$-20 - 20 + \frac{400}{100}$

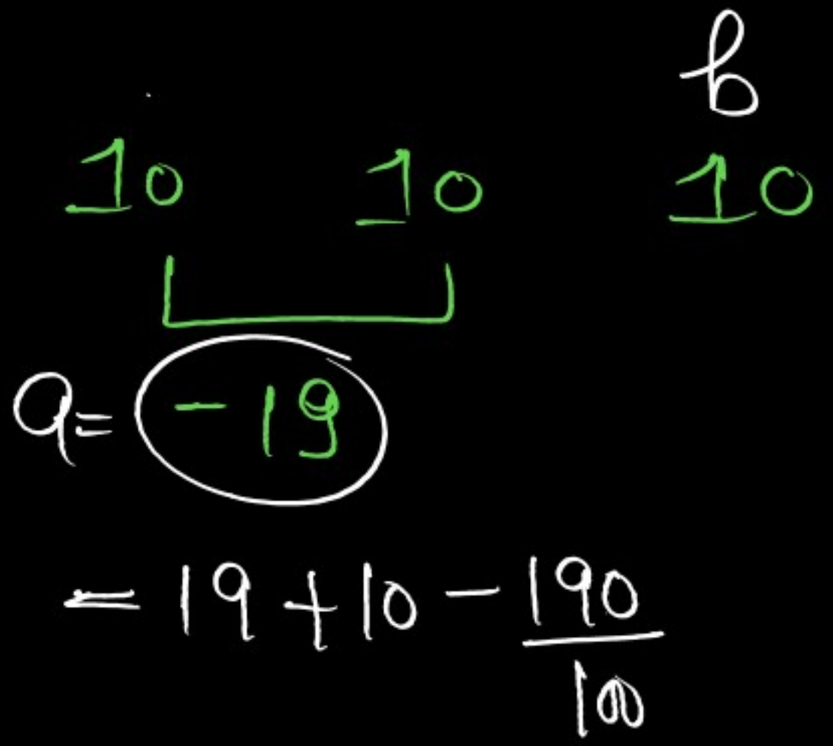
$= -40 + 4 = -36\%$

$20 + 20 - \frac{400}{100}$

$40 - 4$

Question 5:

Three successive discounts of 20% 30% 10% is equivalent to a single discount of:



Question 6:

Which of the following is true: -

a.) buy 2 get 1 free is equivalent to a discount of 50%.: -------

b) buy 3 get 1 free is equivalent to a discount of 25%.: -------

c) buy 2 get 2 free is equivalent to a discount of 40%.: -------

False

33.33%

True

F (50%)

$\frac{\text{Free}}{\text{total}} \times 100$

$\frac{1}{3}$

$\frac{1}{4}$ $\frac{2}{4}$

$\frac{\text{free}}{\text{total}} \times 100$

Question 7:

For any customer which of the following transactions is more profitable?

a) Continuous discount of 20 %and 15%. = 32 %

b) Continuous discount of 10 %and 25%. = 32.5 %

Ans :- c) Continuous discount of 5 %and 30%. = 33.5 %

d) Buy 2 get 1 free? = $\frac{1}{3}$ = 33.33 %

(1) Formulae

$20 + 15 - \frac{300}{100}$

$35 - 3$

(2)

$100 \times \frac{90}{100} \times \frac{75}{100} = 67.5$

(3)

$100 \times \frac{95}{100} \times \frac{70}{100} = 66.5$

Question 8:

The marked price of a grinder was Rs.3,000 and the dealer offered successive discounts of 20% and 15% on it. For how much did he sell the grinder?

C.P ↑ M.P ↓ S.P

M.P ↓

$3000 \times \frac{80}{100} \times \frac{85}{100} = 2040$ Ans

~~680~~ ✗

Question 9:

The marked price of a battery operated rickshaw is Rs.90,000. It is available for sale at two successive discounts of 10% and 5%. What is the selling price (in Rs.) of the battery operated rickshaw?

$90,000 \times \frac{90}{100} \times \frac{95}{100} = 76950$ Am

$10+5-\frac{50}{100} = 15-0.5 = 14.5\%$

Question 10:

Applied to a bill for Rs.1000, what is the difference between a discount of 40% and two successive discounts of 36% and 4%?

$\frac{1.44}{100} \times 1000 = 14.43$

Ans

D = 40 %

36 % 4 %

$36 + 4 - \frac{36 \times 4}{100}$

$= 40 - 1.44$

Diff = 1.44 %

Question 11:

For an electric bill for Rs 50,000, what is the difference between a discount of 30% and two successive discounts of 26% and 5%?

$\frac{3}{10 \times 100} \times 50,000 = 1500$

$D = 30\%$

$|26\% \quad 5\%|$

$= 31 - 1.3$

$= 29.7\%$

$D = 0.3\%$

Question 12:

A shopkeeper wanted to sell Rs.2,000 worth of products. But he had two options, giving three successive discounts of 10% each or giving a single discount of 28%. What was the difference between the two options of discounts (in rupees)? 18 Ans.

10 10 10

$= 19 + 10 - \frac{190}{100}$

$= 29 - 1.9$

$= 27.1\%$

28%

$\frac{0.9}{100} \times 2000$

Question 13:

X and Y Both sell the same product. The marked price of the product is Rs.28000.

X gives a discount of 10% on whole , while Y gives the discount of 12% on the first 20000 & 8 % on the rest of rs8000.what is the difference between their selling price ?

Ans: 240

28000 × 0.9

20,000 × .88 +

10%

D = 2800

20000 | 8000

12%

8%

2400 | 640

3040

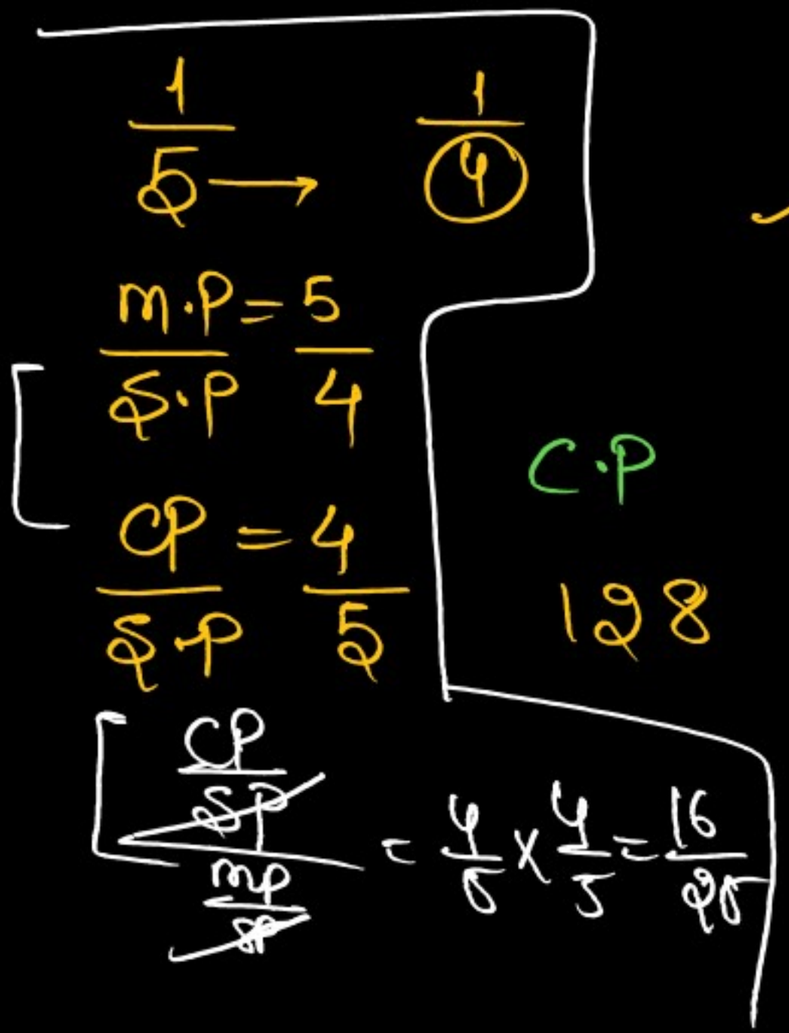
Previous Year Questions

Question 14:

A shopkeeper offers 20% discount and still makes profit of 25%.

Calculate the cost of article which has a marked price of Rs 200.

a) Rs. 130
b) Rs. 128
c) Rs. 125
d) Rs. 140



M.P S.P

200 x 0.8 = 160

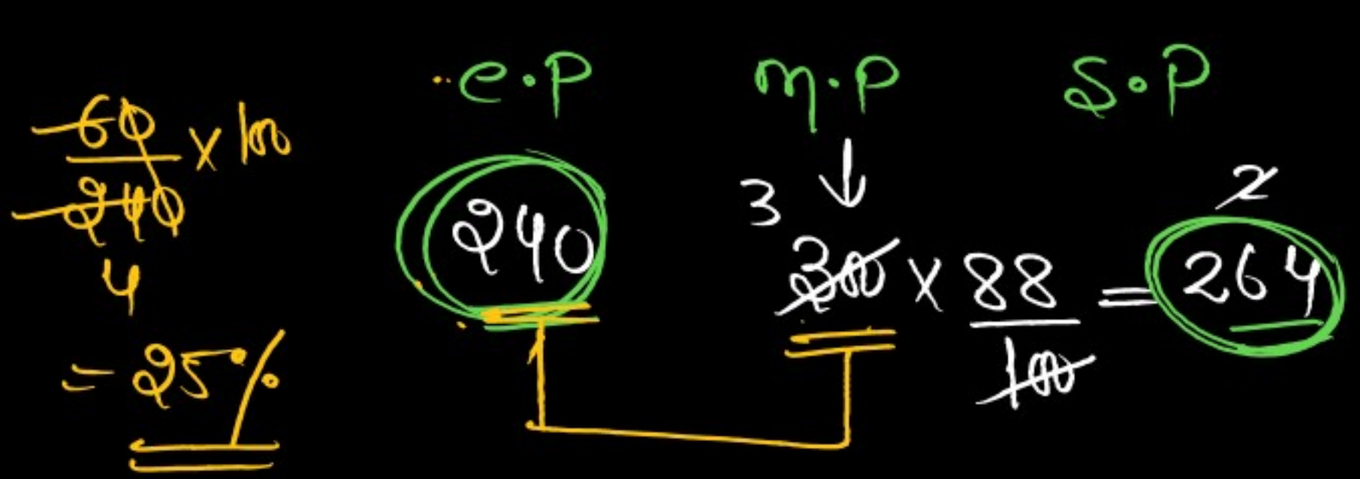
S.P = C.P x M.F

160 = C.P x 5/4

Question 15:

The marked price of an electric iron is Rs.300. The shopkeeper allows a discount of 12% and still gains 10%. If no discount is allowed his gain percentage would have been?

$$24$$
$$264 = CP \times \frac{11}{10}$$



Question 16:

The mark price of an article is Rs. 300.After allowing a discount 15% on the marked price, the shopkeeper makes a gain of Rs.15. What is the gain percentage?

$$\frac{15}{240} \times 100 = 6.25\%$$

CP MP SP

240 $\quad \frac{300}{3} \times \frac{85}{100} = 255$

+15

Question 17:

A shopkeeper marked the price of a jeans 40% above its cost price and gives two successive discounts of $14\frac{2}{7}$ % and 12.5% respectively. If he got total profit of Rs 80, then find the selling price of jeans?

1. Rs. 1680
2. Rs. 1640
3. Rs. 1600
4. Rs. 1620
5. Rs. 1660

5 = 80

1 = 16

CP 100

MP 140

SP $140 \times \frac{6}{7} \times \frac{7}{8} = 105$

$105 \times 16 = 1680$

$14\frac{2}{7}\% = \frac{100}{7} \times \frac{1}{100}$

$\frac{1}{7} = 14.28\%$ | $\frac{1}{14} = 7.14\%$

Question 18:

A shopkeeper marked up price of a laptop 40% above its cost price and gives a discount of $21\frac{3}{7}\%$ and made a profit of Rs. 2250. If shopkeeper had allowed only $\frac{1}{3}$rd of previous discount, then what would have been the selling price of laptop?

(a) Rs. 30500
(b) Rs. 29750
(c) Rs. 29500
(d) Rs. 29250 Ans
(e) Rs. 28250

$20\% = \frac{1}{5}$

$100 \times \frac{4}{5}$

$\frac{150}{7} \times \frac{1}{2} = \frac{3}{14}$

	CP	MP	SP
I	100	140	$140 \times \frac{11}{14} = 110$
II		130	

$140 - 10 = 130$

$10 = 2250$

$11 - 225$

$130 \times 225 = 29250$

Question 19:

Marked price of a bat is Rs 350.A shopkeeper gives a discount of Rs x% and makes a profit of 12.5% If manufacturing cost of bat is Rs 280, then find the discount percent given by shopkeeper on bat?

a) 10%
b) 6%
c) 4%
d) 3%
e) 5%

Ratio Method :

1:3 → [Java]

1:9

Aptitude + DSA

SQL = JPA

$SP = CP \times [100 \pm P/L]$

$SP = MP \times [100 - D\%]$

$$\frac{MP}{CP} = \frac{\frac{SP}{[100 - D]}}{\frac{SP}{100 \pm P/L}} = \frac{100 \pm P/L}{100 - D}$$

$$= \frac{MP}{CP} = \frac{120}{90} = \frac{4}{3} \text{ Ans}$$

*

Question 20:

In a showroom, the marked prices of various items are displayed in such manner that after giving a 10% discount to the customers, the owner of the showroom earns a profit of 20%.

What is the ratio of the marked price and cost price of each item?

CP	MP	SP
100	$\frac{400}{3}$	120

$$\frac{\frac{400}{3}}{100} = \frac{400}{3 \times 100} = \frac{4}{3}$$